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Optimization of CO2 injection and brine production well placement using a genetic algorithm and artificial neural network-based proxy model

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ABSTRACT

CO2 injection during geological CO2 storage (GCS) can result in large pressure build-ups that can limit the injection and reactivate existing faults. The extraction of resident brine from storage formations can alleviate such pressure rises. One critical challenge in applying an extraction well is determining the optimal well locations with reservoir simulation that could be an accurate approach but computationally inefficient. This work proposes the application of an Artificial Neural Network (ANN)-based proxy model in well-placement optimization problems involving pressure management of GCS. The effectiveness of the proposed approach was assessed by optimizing the placement of the CO2 injection and brine extraction wells in the upper aquifer of the Pohang Basin. Several ANN models with different input features were developed for the optimization study, and the best-performing model on the test data was selected. ANN integrated with a genetic algorithm was used to determine the optimal CO2 injection and brine production well locations that maximize the cumulative CO2 injected. Optimization was demonstrated using a full physics reservoir simulator to verify the results. Six hundred and twenty-two simulation runs were needed to reach the optimal solution using the simulator. Instead, with the proposed workflow, comparable results were achieved with only 120 simulation runs, reducing the number of simulation runs by 80.7%. Employing the ANN-based proxy model in the optimization study proved effective with good accuracy by providing a fast approximation to the full physics reservoir model.

1. Introduction

Global CO_2 emissions have increased by approximately 50% since the industrial revolution, reaching the highest average annual atmospheric concentration in 2020 (IEA, 2021). There is now a global consensus that the world should act immediately to prevent the devastating effects of climate change (Metz et al., 2005). Geological CO_2 storage (GCS) in deep formations is a promising tool for tackling climate change (Metz et al., 2005). Saline aquifers, depleted gas and oil reservoirs, and coal bed methane are potential target formations for GCS (Bachu, 2000). Among these geological systems, deep saline aquifers have the greatest potential for GCS (Bachu, 2000).

Although a small fraction of the formation pore space can be used as CO_2 storage, estimates suggest there is sufficient global capacity for GCS as a mitigation option for climate change (Metz et al., 2005). On the other hand, CO_2 storage on a large scale with an injection-only scheme may result in high pore pressure build-up within the formation (Buscheck et al., 2012). This can be severe for formations that represent a closed system since the storage space required in these systems is provided by the compressibility of formation fluid and rock. If pressure build-up is not managed properly, it can limit the dynamic storage capacity and increase the environmental risks (Birkholzer et al., 2012). Therefore, the formation pore pressure should be managed effectively during the injection period for GCS to be a mitigation option for climate warming.

The production of native brine has been proposed as a pressure management tool to effectively control the pore pressure during GCS operations (Bergmo et al., 2011; Lindeberg et al., 2009; Yang, 2008). Brine production can provide many benefits, such as increased storage capacity, reduced risk of environmental impacts, a smaller area of review for monitoring, and steering of the CO_2 plume (Birkholzer et al., 2012; Buscheck et al., 2012). Brine extraction can eventually lead to CO_2 breakthrough in production wells.

A range of studies considered the application of brine production as a pressure management technique and improve storage performance (Bergmo et al., 2011; Birkholzer et al., 2012; Buscheck et al., 2012; Lindeberg et al., 2009; Santibanez-Borda et al., 2019; Yang et al., 2014). Buscheck et al. (2012) examined CO_2 injection combined with brine

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production in synthetic models using vertical and horizontal wells. This study revealed a trade-off between the CO_2 breakthrough time and pressure relief. To manage this, they examined two strategies: a vertical injector with a series of brine producers located incrementally farther to the injectors and horizontal injection/extraction well pairs. With these approaches, they showed that it is feasible to manage pressure relief and CO_2 breakthrough time. Furthermore, their results confirmed that brine production could increase the amount of CO_2 stored 5.5 folds while maintaining the pressure build-up target.

Birkholzer et al. (2012) proposed the concept of "impact-driven pressure management (IDPM)," in which the brine extraction will be targeted to meet the local pressure constraints rather than being equal to the volume of CO_2 injected. The authors assessed the IDPM approach using passive relief wells, active relief wells, and combined passive and active wells. Passive refers to the condition where brine is produced by injection-induced pressure increase in the formation, and active refers to actively pumping brine out. Their findings suggested that implementing active relief wells and combined passive and active relief wells would reduce the extracted brine volume significantly while meeting the criteria for pressure build-up. On the other hand, applying only passive relief wells did not show good results because pressure build-up exceeded the limit.

Cihan et al. (2015) implemented a modified differential evolution algorithm to optimize well placement and brine extraction rates. They employed a numerically vertically averaged two-phase flow model in optimization to make the computational process more efficient. Through this methodology, they estimated the optimal well locations with time-dependent extraction rates to manage the pressure.

Pressure management strategies were also applied in recent CO_2 storage studies in the Pohang Basin. Jun et al. (2019) examined the optimization of well placement and operating conditions for injection only and injection/extraction schemes using different well types and patterns. They showed that placing the producer well at the farthest location closer to the risky zone (faults) resulted in the best combination of pressure management and delayed CO_2 breakthrough. Kim and Shin (2019) proposed a dual tubing injection/extraction scheme using horizontal wells where injection occurs from the heel and production from the toe. With this approach, they managed to improve the performance of the storage system in the Pohang Basin. Hwang et al. (2016) investigated the optimal pressure-relief well design in the Pohang Basin. They optimized parameters, such as the distance between the producer and the injector, brine-pumping rate, and CO_2 injection rate. They reported that storage capacity increased by 45% by producing resident brine compared to the case without production. Han and Kim (2020) studied the spacing between the injector/producer and the length of production interval using a conceptual model representing a shallow layer of the Pohang Basin. They suggested that a shorter production interval placed at the bottom of the reservoir could increase the CO_2 breakthrough time, enhancing the injection performance.

Previous studies suggested that the key challenge in applying relief wells in GCS operation is finding the balance between pressure control and CO_2 breakthrough time. The well locations can have a substantial impact on pressure relief and CO_2 breakthrough time (Buscheck et al., 2012). Therefore, well locations must be designed carefully to obtain the most benefit from the extraction well (Cihan et al., 2015). Most studies found the optimal well locations using a full physics numerical model as the evaluation function (Cameron and Durlofsky, 2012; Hwang et al., 2016; Jun et al., 2019). While this approach provides accurate results, it is computationally exhaustive.

Using numerical models in optimization problems may not be feasible, particularly in well-placement problems. In well-placement problems, the parameter space tends to have multiple local optima because of the heterogeneity and complex reservoir geometry, which may require a large number of forward model runs to obtain a global optimum (Cihan et al., 2015). As a result, computationally efficient methods are required.

Employing proxy models can address this issue. The proxy model (also known as a surrogate model) is an approximation function that can replicate the computationally costly full physics reservoir models. Proxy models, both physics-based (Cardoso and Durlofsky, 2010; Nguyen-Le and Shin, 2022, 2019; Salehian and Çınar, 2019) and data-driven (Golzari et al., 2015; Nguyen-Le et al., 2021; Nwachukwu et al., 2018; Salehian et al., 2022; You et al., 2019), have been widely explored in solving practical problems related to the hydrocarbon industry. On the other hand, in the case of GCS, optimization problems have been conducted mostly using high-fidelity numerical models as the evaluation function.

This paper reports the implementation of an artificial neural network (ANN)-based proxy model in well-placement optimization problems involving pressure management of GCS. To the best of the authors' knowledge, there are no reports on using the ANN model for well placement optimization in the context of GCS. The proposed workflow was tested by optimizing the CO_2 injection and brine extraction well locations in the saline aquifer of the Pohang Basin, Korea. A single realization of the reservoir was employed. The genetic algorithm (GA) is used for both tuning the hyperparameters of the ANN and optimization of well configuration. For the optimization study, a set of ANN models with different input features were developed, and the best-performing model on the test data was selected for deployment. The results from the proposed methodology were verified by comparison with those obtained from the full-physics numerical model as an evaluator.

2. Methodology

This section presents the methodology used for the overall procedure. First, the description of the GCS site and the geological model employed is provided. Next, a description of the optimization algorithms employed is presented. Then, the development of a Proxy model is explained in detail. Lastly, the optimization procedure with various schemes is presented. Fig. 2.1 shows the major steps for the proposed optimization framework.

2.1. Site and geological model description

The saline aquifer in the Pohang Basin investigated in this study is located in southeast Korea (Fig. 2.2). This aquifer was chosen as a small-scale pilot project as part of Korea's effort to reduce emissions (Kim and Shin, 2019). The aquifer bounded with faults outside and within is highly compartmentalized, representing a closed aquifer system. Injection without pressure management in this aquifer can result in high-pressure build-up, which can reactivate pre-existing faults (Kim et al., 2018).

The 3D heterogeneous geological model of the Pohang basin was adapted from previous research by Jun et al. (2019) and Kim and Shin (2019). The depth of the geological interval considered as a target for CO_2 injection ranged from 613 m to 810 m. The Pohang basin has a complex structure with six faults existing within and at the boundary identified through the seismic surveys: western side faults (WF1 and WF2) and eastern side faults (EF1, EF2, EF3, and EF4) (Jun et al., 2019) (Fig. 2.3). It was assumed that exceeding the critical pressure threshold of 13 MPa may reactivate pre-existing faults.

The target CO_2 storage horizon has a continuous low permeable layer that separates the reservoir into upper and lower aquifers (Fig. 2.4). This layer acts as a barrier for vertical flow that is essential to the GCS operation because the CO_2 plume is expected to flow upwards in the vertical direction due to buoyancy. The upper aquifer consists of successive layers of sandstone and mudstone, while the lower aquifer consists of alternating layers of sandstone, mudstone, and conglomerate (Kim et al., 2019). The average horizontal permeability of the upper and lower aquifer is 67.5 and 32.4 md, respectively (Jun et al., 2019). Only the upper aquifer was considered owing to limited computational resources.

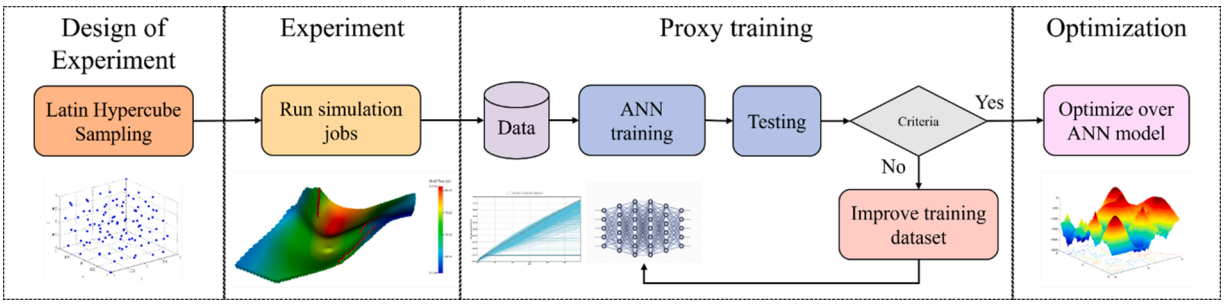


Fig. 2.1. Workflow of the proposed optimization framework.

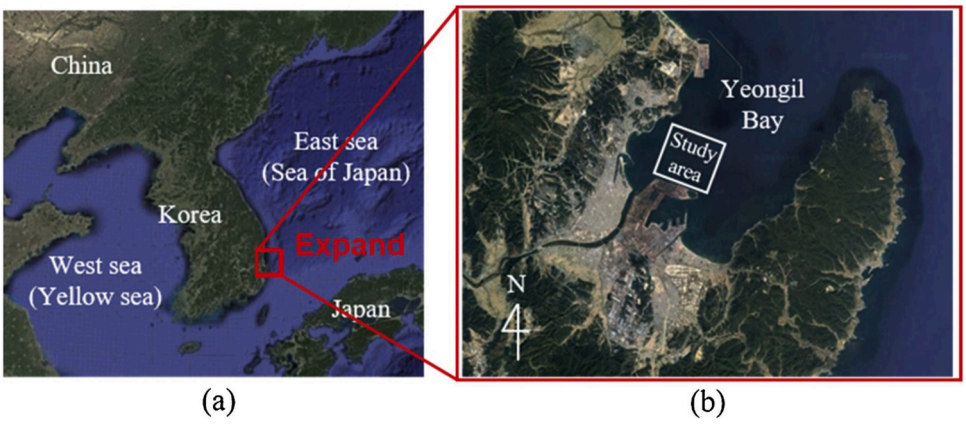


Fig. 2.2. Location of the Pohang basin (a) and GCS pilot project study area (b) (Jun et al., 2019).

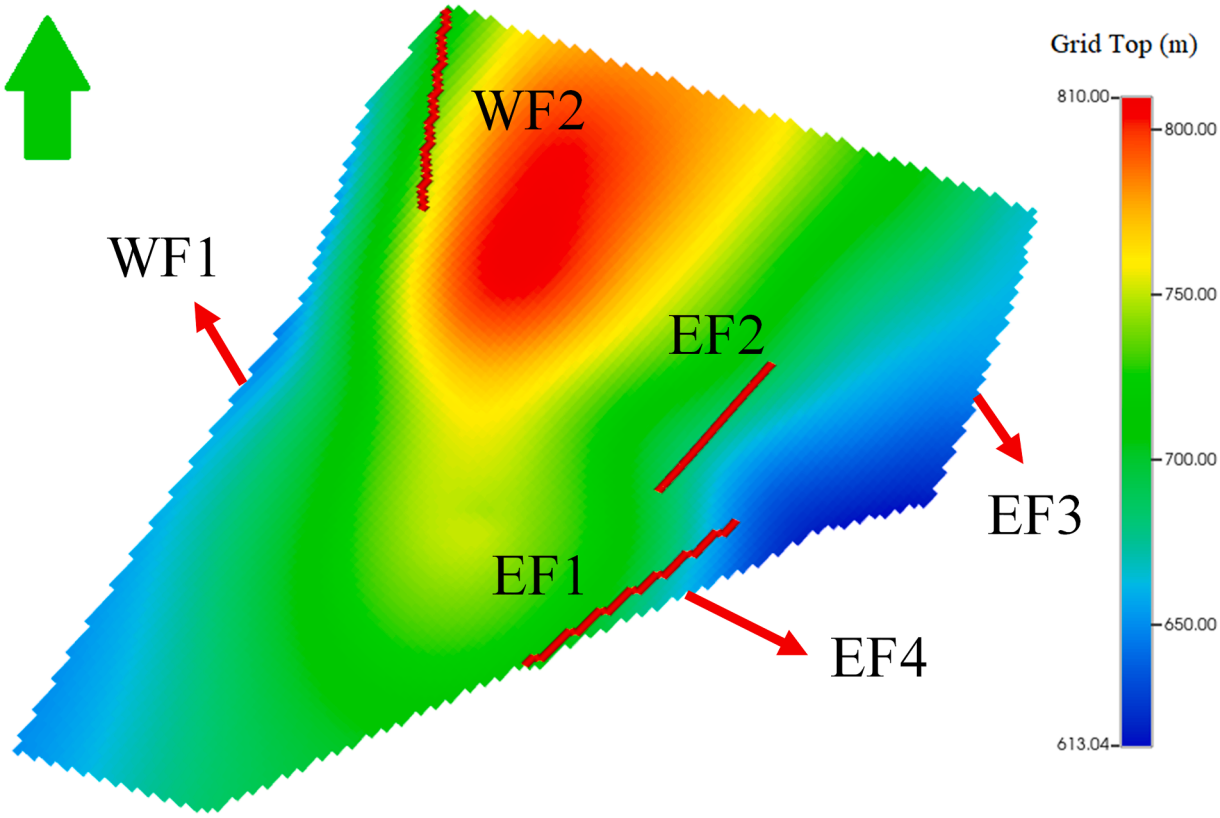


Fig. 2.3. Aerial view of the Pohang basin (depth are shown here). Pre-existing faults within the boundary are identified using red lines.

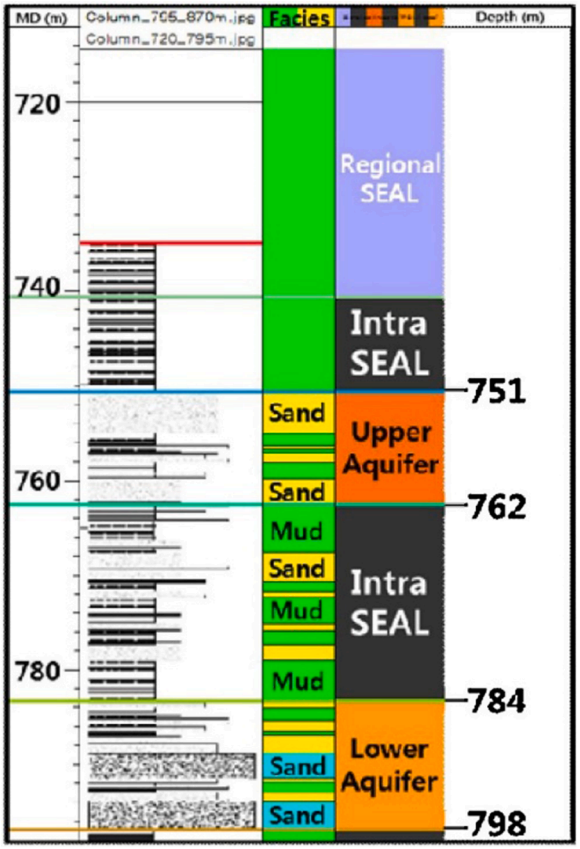


Fig. 2.4. Interpretation of the injection well log data of a small scale pilot project in Pohang Basin (Song et al., 2020).

The extracted sub-model (upper aquifer) domain has a lateral extent of 2.8 × 2.2 km and comprises 95, 152, and 12 grids in the I, J, and K directions, respectively. The 12 grid layers have a total thickness of 12 m. The grid dimensions were 20 × 20 m in the I and J direction, respectively. The eastern and western boundaries were assumed to be closed because they are bounded by sealing faults that impede horizontal flow. Because no geological data were available for the northern and southern boundaries, the no-flow boundary condition was set as a conservative assumption (Kim and Shin, 2019). In addition, the top and bottom of the aquifer were also modeled as a closed boundary. Fig. 2.5 presents the 3D visualization of the reservoir model. Table 2.1 lists the average properties of the upper aquifer. The critical pressure threshold of 13 MPa used in this study was estimated using a semi-analytical solution that takes into account poroelastic stressing caused by subsurface fluid injection (Lee et al., 2017). The initial pressure of the aquifer was estimated using a hydrostatic pressure gradient at the reference depth (Kim et al., 2018). The simulation's relative permeability follows the curves in Fig. 2.6, which are the result of special core analysis on a gas reservoir near to the Pohang Basin in the East Sea (Jun et al., 2019).

2.2. Optimization algorithms

2.2.1. Designed exploration and controlled evolution (DECE)

The DECE algorithm developed by the Computer Modeling Group (CMG) was successfully applied to optimization problems in real fields (CMG, 2021). The algorithm iteratively searches for optimal solutions. As the name suggests, it consists of two stages: (1) Designed Exploration and (2) Controlled Evolution. The algorithm in the Designed Exploration stage randomly seeks the search space to obtain the maximum information. At this stage, different combinations of parameter values (well locations in this study) are generated using the Tabu search techniques

Table 2.1
Parameters considered in the geological model of the upper aquifer, modified from (Jun et al., 2019).

Parameters	Values
Depth of top (m)	746
Thickness (m)	12
Average porosity	0.157
Average horizontal permeability (md)	67.5
kv/kh	0.1
Temperature (°C)	55
Pressure at 750 m (MPa)	7.5
Salinity (ppm)	100,000
Fracture pressure (MPa)	14
Reactivation pressure of faults (MPa)	13

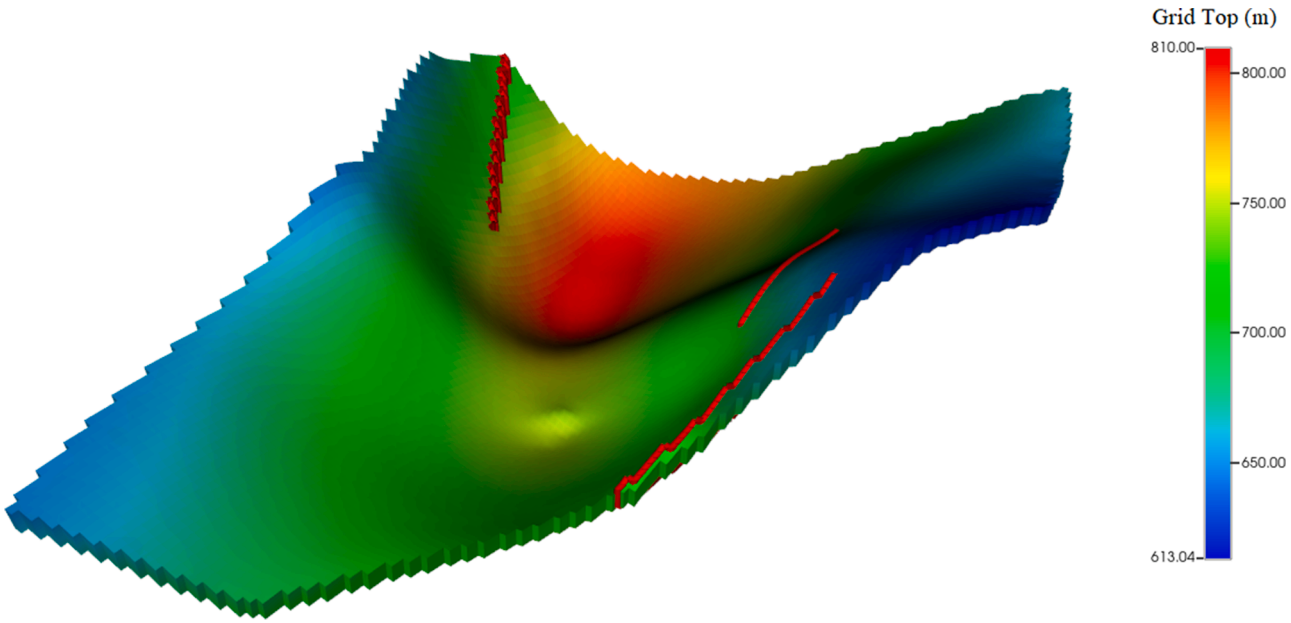


Fig. 2.5. 3D heterogeneous reservoir model of the upper aquifer of the Pohang basin with fault throws shown in red lines (depth shown here).

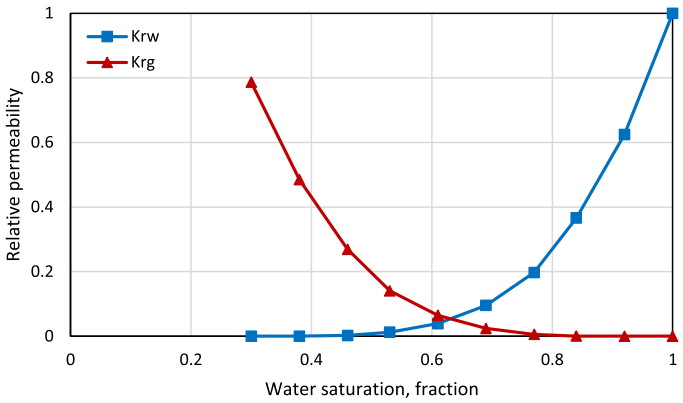


Fig. 2.6. Relative permeability curves for the Pohang geological model simulation (modified from Kim et al., 2018).

and experimental design. In the Controlled Evolution stage, the algorithm decides whether to reject or pick certain candidates through statistical analysis. As the optimization process proceeds to the next iterations, rejected candidates will not be used in exploration. On the other hand, the algorithm will check if the rejection decisions are still valid to avoid trapping in the local optima. If decisions are invalid, those rejected candidates will be recalled and used again in the exploration stage. Fig. 2.7 depicts the workflow of the DECE algorithm.

2.2.2. *Genetic algorithm*

The GA is a nature-inspired stochastic search algorithm developed by Holland (Holland, 1992). By mimicking the natural selection and breeding process, the GA can generate high-quality solutions for optimization problems. The genetic search is conducted over a population of candidates (individuals). The set of individuals forms the current generation. New individuals are produced at each iteration using GA operators, such as selection, crossover, and mutation. At each iteration, individuals are evaluated based on a fitness function. Individuals with high fitness scores are more likely to be selected, reproduced, and represented in the next generation. The algorithm continues to iterate through this process until the stopping criterion is met. Fig. 2.8 shows the main stages of the GA workflow.

2.3. *Development of a proxy model*

An ANN was used to construct a proxy model as a substitute for the full-physics model. The major procedures in structuring the proposed optimization framework are as follows:

a) Initial data preparation: prepare a training database by running the high-fidelity numerical model.
b) Input selection and ANN architecture optimization: develop an ANN model that accurately predicts and generalizes never-before-seen data by choosing the most informative input features with the optimal model hyperparameters.
c) Coupling the ANN model with GA: the developed ANN model integrated with GA is used to perform the well-placement optimization task in the Pohang basin.

2.3.1. *Artificial neural network*

McCulloch and Pitts introduced ANNs inspired by the biological neuron activity in animal brains (Géron, 2019). ANNs are a powerful tool that can map the underlying relationship of input and output data within the boundaries of training data. One common ANN is the Multilayer Perceptron (MLP) network that uses a backpropagation training algorithm in learning (Géron, 2019).

MLP is composed of one input layer, one or more hidden layers, and one output layer (Fig. 2.9). Every layer is fully connected to the next layer and has a weight (w) matrix and a bias (b) vector. The activation function gives MLP the ability to learn complex nonlinear systems. There are many activation functions to consider when training a model, such as the sigmoid logistic function, the hyperbolic tangent function, and the ReLU function.

The model was trained on the training set and evaluated on the testing set to ensure that it was not overfitting to the training data. In addition, early stopping was used during the training process to prevent the model from continuing to learn from the training data after it had reached its optimal performance. Specifically, the validation loss was monitored during training and stopped the training process when the validation loss stopped improving. Moreover, L2 regularization technique was employed in the model where it adds a penalty term to the loss function that encourages the model to use smaller weights, which can help prevent the model from overfitting to the training data.

2.3.2. *Data preparation*

The first part of developing a proxy model is data preparation. The quality of the input data significantly impacts the accuracy of the resulting proxy model. The training data that covers the wide range in the design space can enhance the performance of the proxy model (Golzari et al., 2015). Thus, Latin Hypercube Sampling (LHS) was utilized to prevent uneven sampling within the design space. Initially, 100 samples (a combination of well locations) were obtained in a space-filling manner using LHS.

The response variable selected for this study is cumulative CO_2 injected. The response variables for the initial training dataset were obtained by running 100 samples (equal to 100 simulation runs) in a numerical simulation. This study used the numerical simulation software GEM from CMG. The simulation outputs (cumulative CO_2) were reported at 71 timesteps (monthly intervals) ranging from 31 to 2161 days. All the input data were normalized to values between 0 and 1 using min-max scaling before being fed into the ANN.

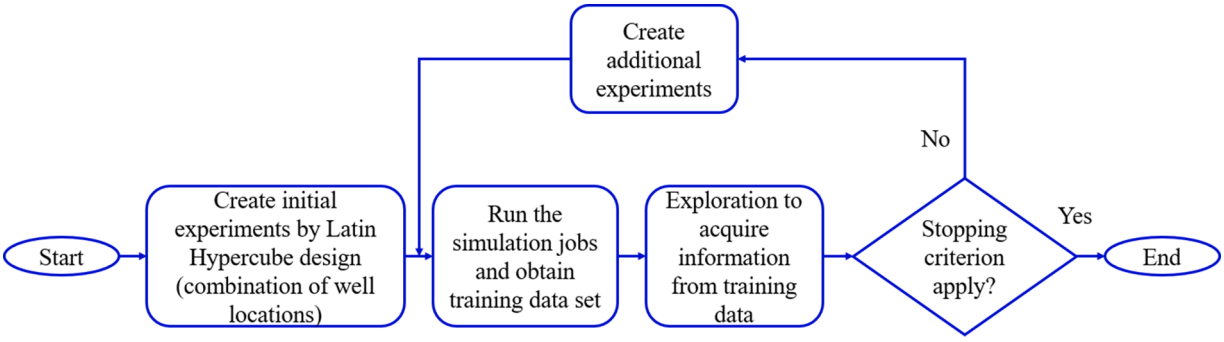


Fig. 2.7. Workflow of the DECE algorithm optimization process.

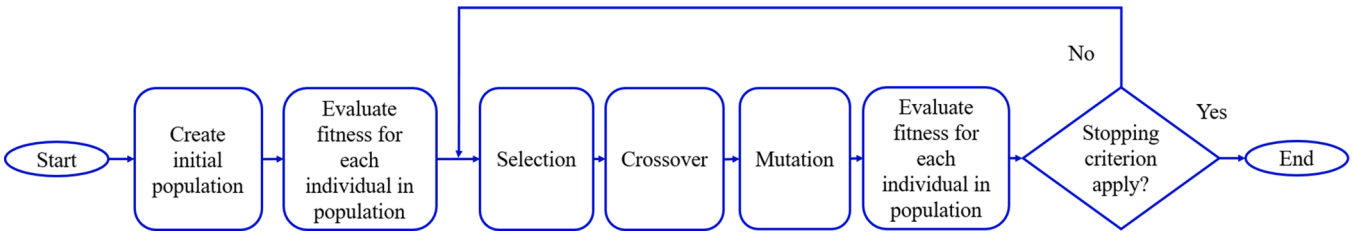


Fig. 2.8. Workflow of the GA optimization process modified from (Wirsansky, 2020).

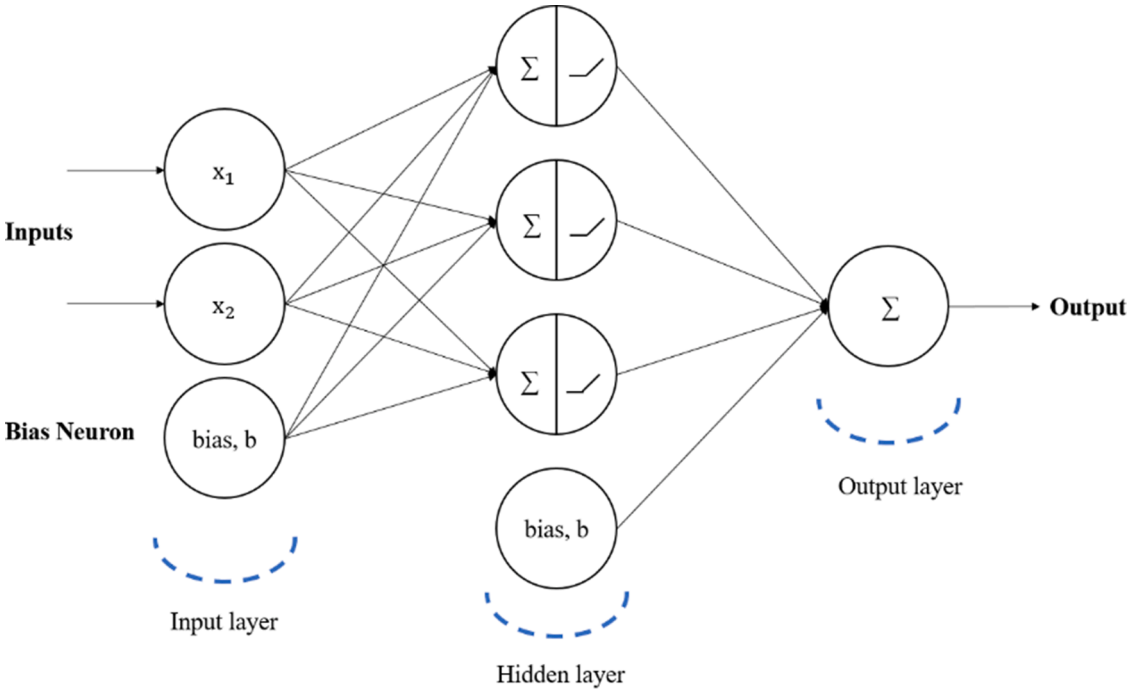


Fig. 2.9. Schematic diagram of a Multilayer perceptron with one hidden layer.

2.3.3. Input selection and hyperparameter tuning

When developing an ANN model, it is a good practice to select the most informative input features that will help the model generalize better to the never-before-seen data (Chollet, 2017). Therefore, different models were developed (Base, Model A, Model B, and Model C) with different input features and trained using initial training data. The goal was to select the model that performed best on the test data.

In the Base Model, only injection and production well-gridblock addresses (I, J) were considered the input along with the elapsed time, which is the durated time since the CO2 injection is started. In Model A, in addition to well-gridblock addresses and the elapsed time, the well (both the injector and producer) block permeabilities and porosities, inter-well distance, and distance from the injection well to the closest boundary were considered (Centilmen et al., 1999; Nwachukwu, 2018). In Model B, all the features were identical to Model A except for the well-gridblock permeabilities and porosities. Instead, the productivity potential of the well gridblocks was used.

Productivity potential (PP) proposed by Guerra and Narayanasamy (2006) is a heuristic approach in which a combination of petrophysical properties is represented using a proxy function. The original form of the proxy function includes the saturation of oil which was removed since the focus area is a saline aquifer (Eq. (2.1)). PP is given by the following:

$$PP = \sum_{i=1}^{n} k_i \varnothing_i \tag{2.1}$$

where n is the total number of vertical layers; k_i and $\varnothing_i$ are the permeability and porosity at the ith grid, respectively.

Min et al. (2011) proposed the usage of productivity potential as an input feature to the ANN model. In their study, they showed that employing PP improved the accuracy of the ANN model in predicting cumulative oil production. Thus, in Model B, the productivity potential was considered an input.

The final model considered was Model C. All the input features were

Table 2.2
Input features of the developed models.

Base model	Model A	Model B	Model C
Injector gridblock address (I, J) Producer gridblock address (I, J) Elapsed time	Injector gridblock address (I, J) Producer gridblock address (I, J) Injection well block porosities& permeabilities Production well block porosities& permeabilities Inter well distance Distance from the injection well to the closest boundary Elapsed time	Injector gridblock address (I, J) Producer gridblock address (I, J) PP of well gridblock (injector& producer) Inter well distance Distance from the injection well to the closest boundary Elapsed time	Injector gridblock address (I, J) Producer gridblock address (I, J) Average permeability & porosity along the injector Average permeability & porosity along the producer Inter well distance Distance from the injection well to the closest boundary Elapsed time

the same as Model A except for the well-block permeabilities and porosities. Instead, they were replaced with the arithmetic average of permeabilities and porosities along the well blocks (Onwunalu, 2006). The proposed models are shown in Table 2.2.

Hyperparameter optimization was conducted for each model using the GA. The k-fold cross-validation (four-fold) average mean absolute error (MAE) was used as an objective function, whereas ANN models were trained using mean squared error (MSE) as a loss function. MSE is a commonly used loss function in regression problems (Chollet, 2017).

The GA parameters used in the ANN hyperparameter tuning are shown in Table 2.3. The GA parameter values were determined by trial-and-error, taking into account the complexity of the problem, the intended balance between exploration and exploitation, and the tradeoff between computational cost and solution quality (Wirsansky, 2020). It was determined that a reduced population size and fewer generations would be sufficient to explore the search space for the ANN tuning problem. Additionally, a reduced population size can reduce computational costs and potentially accelerate the optimization process. The probability of crossover encourages exploitation, while the probability of mutation promotes population diversity. The number of elites of 2 and a tournament size of 2 were chosen to achieve a balance between exploration and exploitation and preserve population diversity.

The activation function and optimizer were pre-set to ReLU and Adam, respectively. A large number of epochs during the tuning process can be computationally exhaustive; thus, 20 epochs were considered in the tuning process. The number of hidden layers and the number of neurons per layer were optimized by keeping the batch size and learning rate at the default values of 32 and 1e-3, respectively. Table 2.4 lists the hyperparameters used in the tuning process and their corresponding range.

After tuning the hyperparameters, each model was retrained by a larger number of epochs. The epoch numbers were determined by monitoring the cross-validation error. Once the models are trained, they must be evaluated using new data (Chollet, 2017). Thus, 20 test data were generated using LHS corresponding to 20 simulation runs.

The prediction quality of each model in the testing data was assessed using metrics, such as R^2, absolute relative error (E), mean absolute error (MAE), and mean absolute percentage error (MAPE).

The R^2 value is a direct measure of the goodness of fit between predicted and actual values and can be computed by the following formula:

$$R^2 = 1 - \frac{\sum_i^n (y_i - \widehat{y}_i)^2}{\sum_i^n (y_i - \overline{y}_i)^2} \tag{2.2}$$

where n is the number of trained data; y_i is simulation responses; $\widehat{y}_i$ is the proxy predictions; $\overline{y}$ is the mean of the simulation responses (true outputs).

The absolute relative error (E) is the ratio of the difference between the network's actual output and observation targets to the observation targets. It can be computed by

$$E_i = \frac{|y_i - \widehat{y}_i|}{y_i} \tag{2.3}$$

Mean absolute error (MAE) is the arithmetic average of the absolute errors given by:

Table 2.3
GA parameters used in hyperparameter tuning.

Parameter	Value/ setting
Population size	30
Crossover's probability	90%
Mutation's probability	30%
Max. number of generations	50
Number of elites	2
Selection type	Tournament (size 2)

Table 2.4
Hyperparameters and their search range.

Parameter	Range
Number of hidden layers	1–5
Number of neurons	10–100

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| \tag{2.4}$$

The mean absolute percentage error (MAPE) is another measure of the prediction accuracy given by

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right| \tag{2.5}$$

2.4. Optimization problem formulation

The main goal in optimization was to find the optimal well locations that maximize the amount of cumulative injected CO_2 at the end of the injection period (six years). CO_2 injection and brine extraction wells were considered. Only vertical wells were used. The optimization problem, including the objective functions and constraints, can be formally expressed as follows (Cihan et al., 2015):

$$\text{Maximize } f(\boldsymbol{x}) = \sum_{k=1}^{n} Q_{k,\ CO_2}\Delta t_k \tag{2.6}$$

$$\text{Subject to } g_1(\boldsymbol{x}) = V_{ext,\ CO_2} = 0 \ \textit{and}\ g_2(\boldsymbol{x}) = \Delta P_{crt} - \max\left\{\Delta P\left(i_{fault}, j_{fault}, t\right)\right\} \rangle 0 \tag{2.7}$$

where $\boldsymbol{x}$ is the vector of control variables that are locations of injection well (i_{inj}, j_{inj}) and extraction well (i_{ext}, j_{ext}); k is the simulation step; n is the total number of simulation timesteps; Δt_k is the length of the kth timestep; $Q_{k,\ CO_2}$ is the injection rate at timestep k; and ΔP_{crt} is the difference between the fault reactivation pressure and initial pressure of the reservoir.

The first constraint in (Eq. (2.7)) ensures that no CO_2 will be extracted from the production well. The Pohang basin contains three faults within the boundary prone to reactivation if the pressure exceeds the critical pressure threshold (see Section 2.1). Thus, the second constraint will keep the pressure in the faults below the critical pressure threshold (ΔP_{crt}).

2.4.1. Feasible locations

The Pohang basin is highly irregular, containing pre-existing faults within the domain. To avoid the potential risks associated with positioning wells near existing faults, a pre-screening criterion was implemented that excluded any well locations within 100 m of the faults. This criterion was predicated on a 13 MPa critical pressure threshold, above which pre-existing faults may become reactivated. Exceeding this threshold could result in potential hazards, so it is essential to ensure that well locations are not in close proximity to fault zones. All optimization models created for this investigation were subjected to the pre-screening criterion. The optimal well locations determined in the models were then evaluated to ensure that they satisfied the pre-screening criteria. This method guaranteed the security and reliability of the optimal well placements for CO2 injection and brine production in the saline aquifer of the Pohang basin. Fig. 2.10 presents the feasible and infeasible locations.

2.5. Optimization schemes

After selecting the best-performing model among the proposed models, it was used in the optimization workflow. The GA was coupled

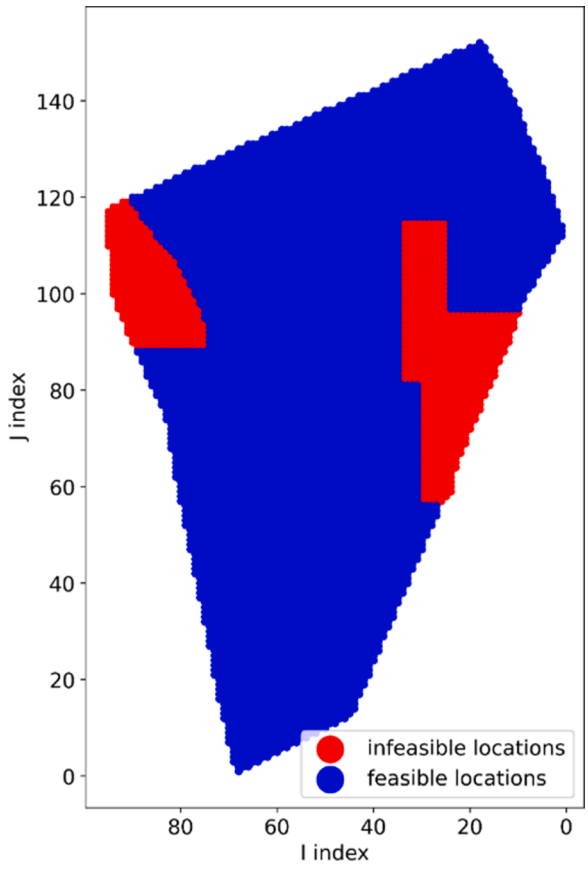


Fig. 2.10. Feasible (blue) and infeasible (red) locations of the Pohang basin considered in this study.

Table 2.5
GA parameters used in the well-placement optimization process.

Parameter	Value/ setting
Population size	200
Crossover's probability	90%
Mutation's probability	30%
Max. number of generations	500
Number of elites	3
Selection type	Tournament (size 2)

with the proxy model to conduct the optimization process. Table 2.5 displays the GA parameters used to optimize well location. The GA parameter values were determined by trial and error. The determined parameters reflected the increased complexity of the well placement optimization problem. A larger population size was chosen to explore wide range of possible solutions, which is essential for difficult optimization problems. The number of generations was determined based on a desired balance between computational expense and solution quality. The number of elites of 3 was selected to maintain the best solutions while allowing for a diverse population.

To compare the results and verify the proposed framework, the optimization task was also conducted employing only the full-physics numerical model. Optimization was carried out using the intelligent optimization tool CMOST (developed by CMG). The CMOST built-in algorithm DECE (see Section 2.2.1 for an explanation) was used as an optimizer.

The summary of all optimization schemes is as follows:

1 Case 1: a classic approach where optimization was conducted using a high-fidelity reservoir model.
2 Case 2: optimization was conducted using only the proxy model. Once the proxy was constructed with a proposed workflow simulator went offline, further updates of the proxy model were not performed. The optimal solution found was evaluated using the numerical simulator and accepted as an optimal solution (or near optimal).
3 Case 3: Optimization was conducted using the proxy model updated during the optimization process. This is also known as an online strategy in the literature, which was proven to improve the accuracy of optimal solutions (Golzari et al., 2015; Salehian et al., 2022). Different authors used different approaches to update the proxy model in the literature. In this case, the optimization was carried out first using the model trained with initial training data. Next, the best solutions (combination of well locations) obtained were evaluated using the reservoir simulator and added to the training dataset. The proxy model was retrained using updated training data. Subsequently, the updated proxy model was used for the next optimization run. This process was repeated until the model showed satisfactory performance.

3. Results

3.1. Results from the hyperparameter tuning procedure

As described in Section 2.3.3, four different models with different input features were developed. The input features were selected based on previous studies.

Fig. 3.1 shows the progression of the GA hyperparameter tuning process for all models. The tuning procedure improved the hyperparameters compared to the initial generation. Model B reached the lowest cross-validation error during the tuning process. At this point, however, a conclusion about the performances of the models cannot be drawn because fewer epochs (20) were used during optimization.

Table 3.1 lists the optimized hyperparameters for all considered models. The number of values in the brace represents the number of hidden layers used in each architecture, and the values represent the number of neurons in each hidden layer. For example, the Base Model with the {96, 24, 90, 67, 60} architecture represents an architecture of five hidden layers with 96, 24, 90, 67, and 60 neurons in each. Differences were observed in the tuned architectures of each model,

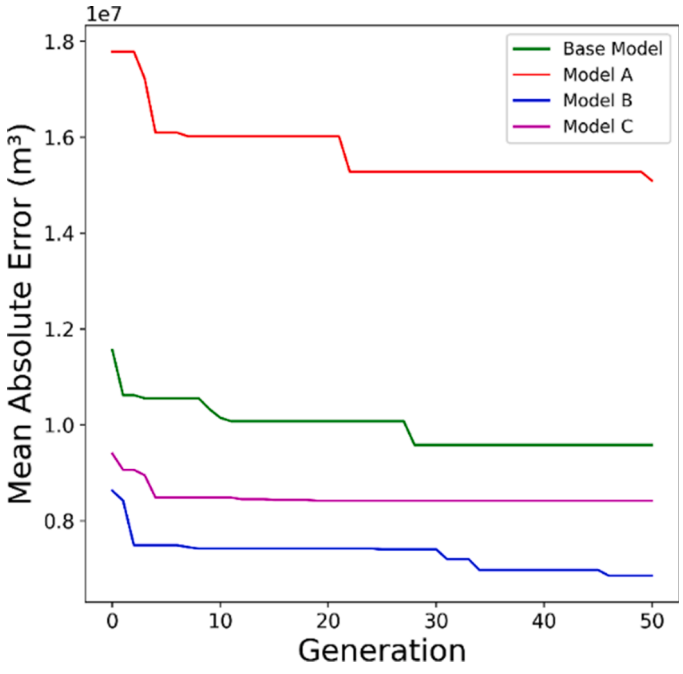


Fig. 3.1. Hyperparameter optimization progress for the four models.

Table 3.1
Tuned architecture of each model.

Model	Architecture
Base Model	{96, 24, 90, 67, 60}
Model A	{99, 73, 63, 10, 41}
Model B	{100, 86, 100, 11, 94}
Model C	{99, 69, 27, 60, 71}

suggesting differences in the complexity of models in predicting the output.

Using the optimal ANN architectures, all models were retrained with a higher number of epochs. The optimal number of epochs was determined by monitoring four-fold cross-validation MAE. The performances of the models on the test data are given in Fig. 3.2. Generally, a higher correlation factor (R^2), lower MAPE, lower absolute relative error, and a lower MAE correspond to higher predictive model accuracy.

The Base Model represents the case where elapsed time and well spatial locations were used as input. Although the performance of the Base model was acceptable, it was not accurate. Because the Pohang basin is heterogeneous, the Base Model could not predict the cumulative CO_2 accurately (Fig. 3.2 and Table 3.2). Hence, the predictive model that is only trained using well locations is incapable of predicting the behavior of heterogeneous formation (Nwachukwu, 2018). This suggests that the inclusion of features that are influential on injection and production behavior is required to build a model that accurately predicts.

In Model A, the well-block permeabilities and porosities were input individually to capture the heterogeneity effects. It is the worst performing model among all models. Although the permeability and porosity of the well blocks provide information on the reservoir properties, they increase the dimensionality of the input features. Because the upper aquifer of the Pohang basin consists of 12 layers, adding permeability and porosity of the well blocks as inputs in Model A increased the number of input features to 55 (Table 3.2). Increasing the number of input features in machine learning problems can make it difficult to find a solution, which is referred to as the curse of dimensionality (Géron, 2019). Thus, high dimensionality can be one of the reasons for the low performance.

Models B and C are the best performing models (Fig. 3.2 and Table 3.2). Model B performed slightly better than Model C. The reason why Model B and C outperformed other models can be explained by the significant decrease in the dimensionality of input variables while including the most informative features (e.g., permeability and porosity) (Min et al., 2011).

The performance of Models B and C was deemed satisfactory, and both can be selected for deployment. Model B was chosen over C to conduct an optimization study owing to its slightly higher performance. For illustration purposes, the optimization using the Base and Model C was also conducted. Table 3.2 provides a summary of the performances of all models.

3.2. Optimization results

Well-placement optimization was investigated in the upper aquifer of the Pohang basin. In the process of optimization, two vertical wells were utilized, one for CO2 injection and the other for brine production. The injector operated by a constant bottom hole pressure of 13.3 MPa, and the producer operated by a constant bottom hole pressure of 7.5 MPa. The producer will shut in if CO_2 breaks through. As described in detail in Section 2.5, three approaches were explored in the optimization task:

- Case 1: optimization using only the numerical simulator (CMG-GEM). DECE algorithm was employed as an optimizer.
- Case 2: optimization using only proxy model (ANN). The GA was employed as an optimizer.
- Case 3: optimization using proxy model (ANN) updated during the process. The GA was employed as an optimizer.

In all cases, the optimization was repeated several times with different random seed numbers. Only the best results are presented here.

3.2.1. Case 1

Initially, optimization was conducted using only a numerical simulator. The CMOST intelligent tool and built-in optimizer DECE were used. The optimization progress is shown in Fig. 3.3. Optimization was conducted for 1000 experiments, and the optimal solution was reached after 622 simulation runs.

Fig. 3.4 shows the optimal well locations. Intuitively, the injection well was expected to be farther away from the fault zones to ensure that the fault pressure remained below the critical threshold. As shown in Fig. 3.4, the injection well location was farther from the fault zones (red). However, it revealed that the injection well is located close to the west boundary, which increases the possibility of the CO2 plume reaching the boundary. No-flow conditions were assumed at boundaries throughout simulation. As a result, the boundaries prevent the CO2 from escaping. This makes sure that, regardless of whether the CO2 plume reaches the boundary or not, the simulation appropriately represents the migration of the plume in the actual field condition. On the other hand, brine extraction was closer to the fault zones because it helped relieve the pressure.

3.2.2. Case 2

In this case, the proxy model (Model B) was trained initially with 100 simulation runs. Later, the simulation went offline, and optimization was conducted without further updates on the constructed model.

Fig. 3.5 shows the cumulative CO_2 of the best well locations obtained at each generation evaluated using the proxy model (solid blue line) and actual numerical simulation (solid red line). Here the actual numerical simulation for the best well locations in each generation is shown for illustration purposes only. Deviations from the actual reservoir output were observed in most generations because of the low performance of the ANN model in newly observed areas in the search space discovered by the GA. Because the model was built using only 100 simulation runs, it is a rough approximation of the full physics model.

Table 3.3 presents the optimization results from both full-physics simulator and proxy models. "Error" refers to the percentage difference between the cumulative CO2 amount obtained from the proxy model and the one obtained from the full-physics simulator for the initial and optimal solutions. "Change" refers to the percentage change in the cumulative CO2 amount between the initial and optimal solutions. The optimization results showed that the optimization converged to a better solution compared to the initial starting point, and close agreement between the ANN prediction and simulation output was observed for the optimized configuration.

Another point to notice from Fig. 3.5 is the number of objective function evaluations. With an individual size of 200 and 500 generations, 100,000 forward runs were performed during the GA optimization process, which would be impractical without the fast proxy model. Fig. 3.6 shows the optimal well locations obtained by Case 2.

3.2.3. Case 3

In Case 3, the proxy model was updated twice during optimization. Eleven best solutions obtained from the first optimization using the initial model were evaluated in the simulator and added to the training data to update the model for the first time. The 10 best solutions obtained from the second optimization run using the first updated model were then added to the training data to update the model for the second time. After the second update, the model performance was deemed satisfactory. The final model used in the optimization process was

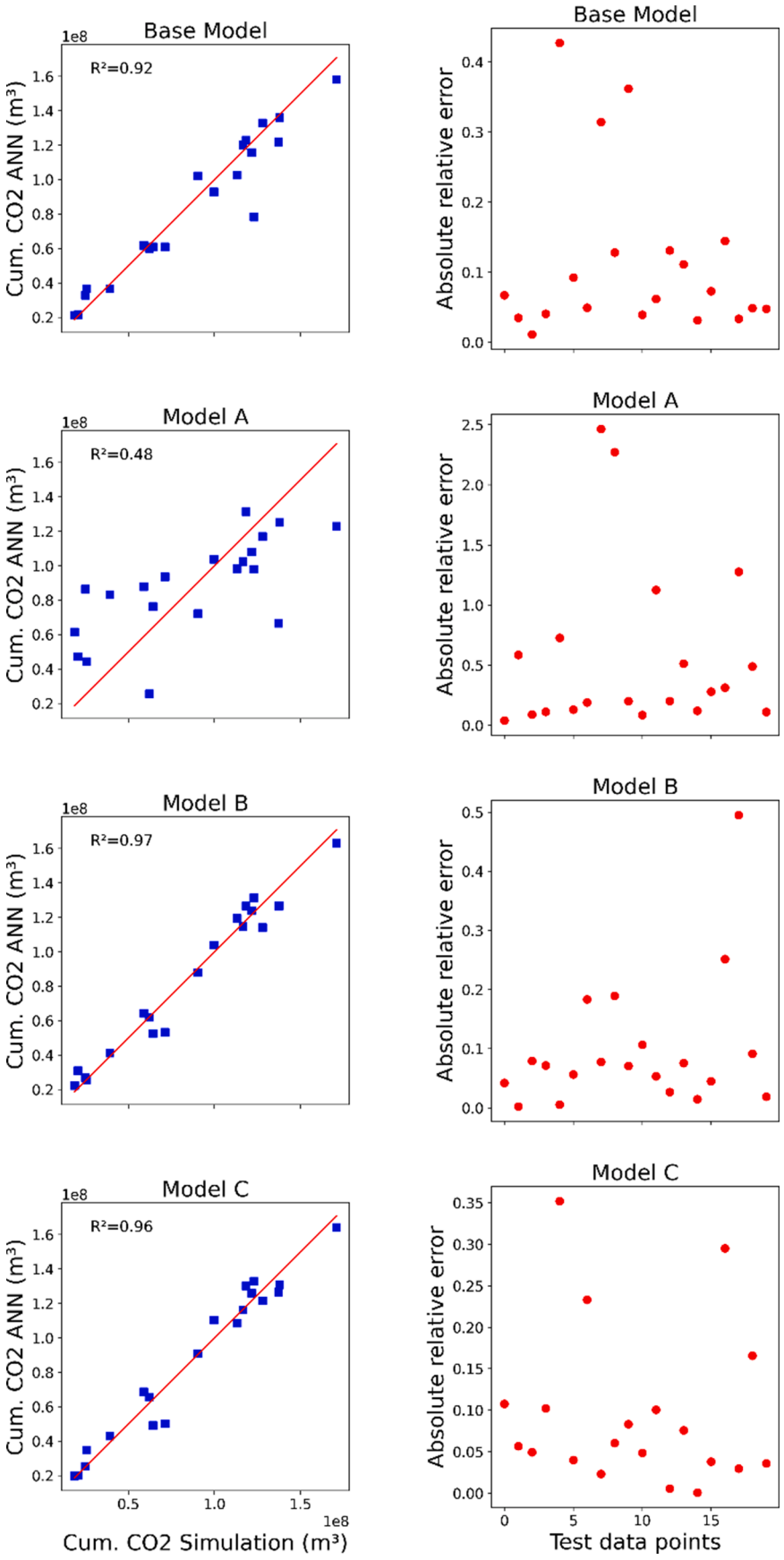


Fig. 3.2. Performance of the models in testing data (total 20). (Col. 1) Scatter plot of true and predicted cumulative CO_2 stored at the end of the injection period (six years). (Col. 2) Absolute relative error for the test data points.

Table 3.2
Summary of the performances of the models on test data and number of input features used in constructing each Model.

	Number of inputs	R^2	MAPE	MAE (m^3)
Base	5	0.92	11.3%	8.265E + 06
Model A	55	0.48	56.7%	2.691E + 07
Model B	9	0.97	9.8%	6.518E + 06
Model C	11	0.96	9.5%	6.858E + 06

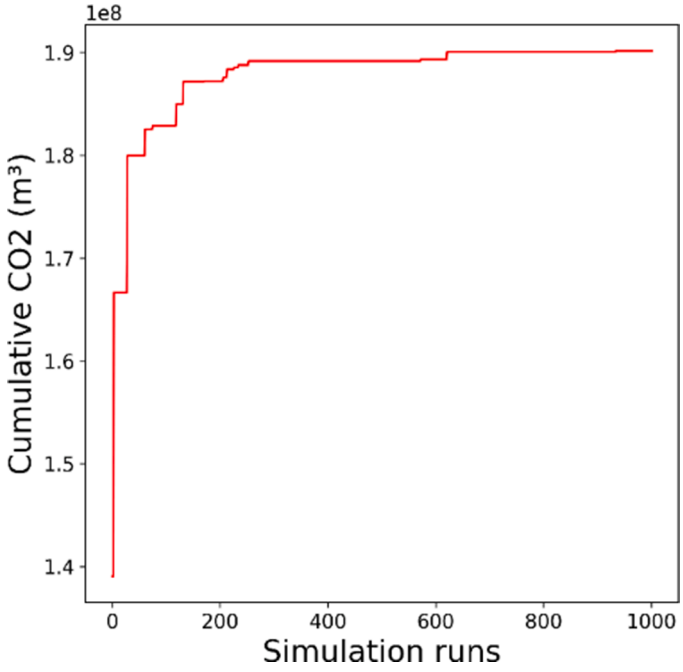


Fig. 3.3. Optimization progress of Case 1.

trained using 121 simulation runs (including the 100 initial runs).

Fig. 3.7 shows the GA optimization progress. The simulation results of the best well locations in each generation are plotted for illustration purposes. The ANN prediction improved compared to Case 2 as the difference between the ANN prediction and simulation output narrowed. The ANN and simulation outputs show excellent agreement with less than 1% error for the optimal well configuration, as shown in Table 3.4. Fig. 3.8 shows the optimized well locations.

Both Cases 2 and 3 resulted in similar well locations as Case 1. The best result from the considered proxy model-based optimization scheme was obtained from Case 3. The accuracy of Case 3 outperformed Case 2 because it was retrained by adding near-optimal solutions to the training data. Retraining the model using the updated training data improved its accuracy around the optimal solution, and the model accuracy improved. This finding is in good agreement with studies, such as (Golzari et al., 2015; Jang et al., 2018; Salehian et al., 2022), where authors showed that updating the proxy model during optimization improves its accuracy around the optimal points. Table 3.5 lists all schemes with the optimal well locations and corresponding cumulative CO_2 amount. The number of simulation runs shown in Table 3.5 for Cases 2 and 3 included the 20 simulation runs required to generate test data.

3.3. Computational time

The training cost of ANN is negligible compared to the simulation runtime. Therefore, the computational time was compared based solely on the number of simulation runs required for each case to reach the optimal solution.

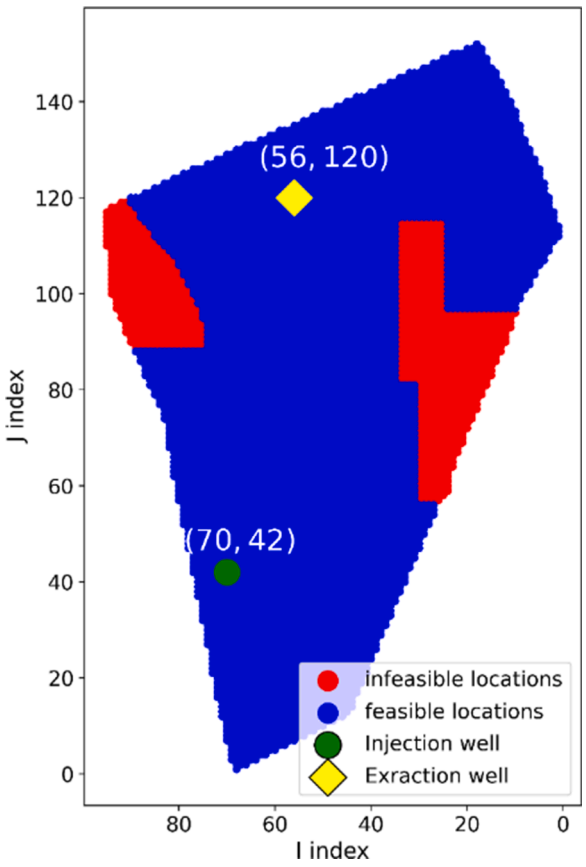


Fig. 3.4. Optimal well locations obtained by Case 1.

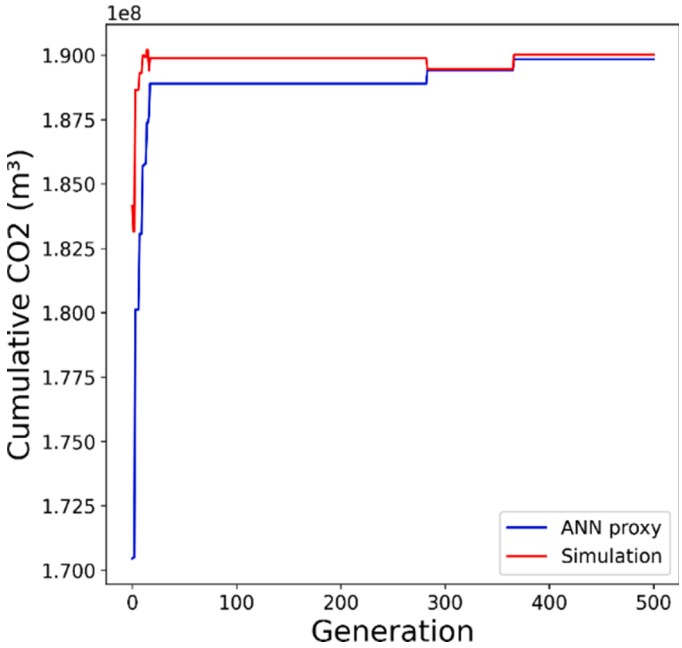


Fig. 3.5. GA optimization progress of Case 2.

Case 1 required 622 simulations to reach the optimal solutions. In contrast, Cases 2 and 3 reduced the number of simulation runs compared to Case 1. The decrease in the number of simulation runs for Case 2 was (622–120)/622 = 80.7%, and for Case 3 was (622–141)/622 = 77.3%. This shows the advantage of applying proxy models with reasonable accuracy and computational efficiency. The proposed approach thus

Table 3.3
Cumulative CO_2 for the initial and optimal solutions obtained by Case 2.

	Proxy model	Full-physics	Error
Initial (1E + 8 m^3)	1.7045	1.8415	7.44%
Optimal (1E + 8 m^3)	1.8985	1.9002	0.09%
Change	11.38%	3.19%	

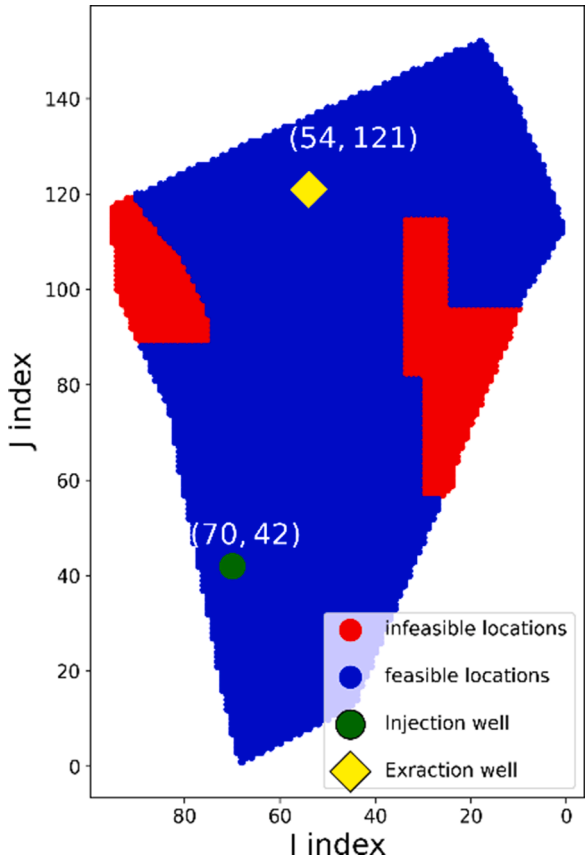


Fig. 3.6. Optimal well locations obtained by Case 2.

appears to be viable for the well-placement optimization task in GCS projects.

3.4. Influence of input feature in optimization results

From the input selection study, increasing the dimensionality (Model A) or building a model with less informative features (Base Model) resulted in lower performance (Table 3.2). Optimization was performed using the Base Model and Model C to demonstrate the importance of feature selection in the optimization problem and compared with the Case 2 (Model B) results. Model A was not included in the optimization because of the low accuracy (Fig. 3.2).

The Base Model, which was constructed using only elapsed time and spatial information of wells, resulted in sub-optimal conditions (Table 3.6). The observations agree with authors who also indicated that the model constructed using merely spatial locations as an input variable is incapable of finding the optimal solutions for well placement problems (Centilmen et al., 1999; Nwachukwu, 2018). Model C, however, resulted in slightly lower cumulative CO_2 than Model B, suggesting that it can capture the heterogeneity effects because average permeability and porosity were part of the input data.

Fig. 3.9 shows the optimal well locations obtained by each model. The optimal locations of the injection and extraction wells obtained from each model were placed in the same regions. However, differences

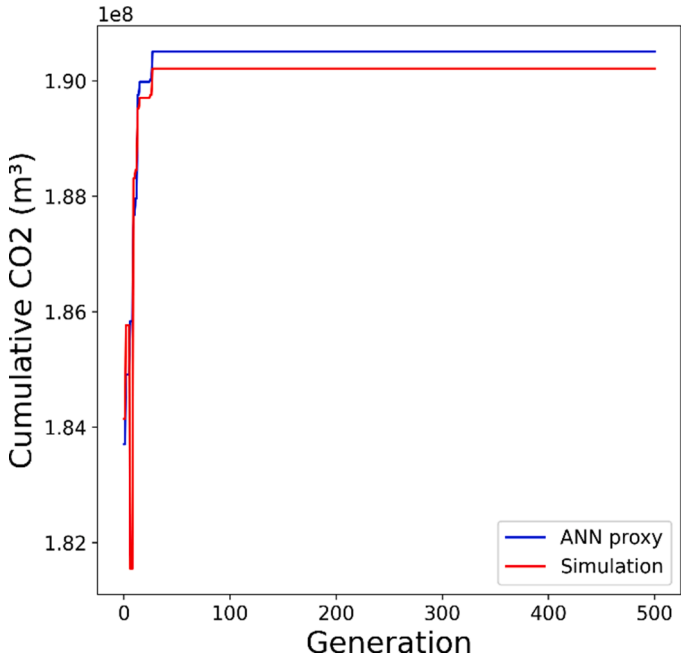


Fig. 3.7. GA optimization progress of Case 3.

Table 3.4
Cumulative CO_2 for initial and optimal solutions obtained by Case 3.

	Proxy model	Full-Physics	Error
Initial (1E + 8 m^3)	1.8370	1.8415	0.24%
Optimal (1E + 8 m^3)	1.9051	1.9021	0.15%
Change	3.70%	3.30%	

were observed in the distances between the injector and the producer.

The challenge in applying the pressure management well in GCS is finding the balance between increased injectivity and CO_2 breakthrough time (Buscheck et al., 2012). A higher injection rate can be achieved by placing the extraction well closer to the injection well. On the other hand, if the distance is too close, it results in early CO_2 breakthrough and lowers the cumulative CO_2 stored. Although the CO_2 breakthrough in the Base model occurred somewhat later than in the other models, the lower injection rate resulted in lower cumulative CO_2 stored. On the other hand, optimal solutions obtained by Models B and C achieved a better trade-off between the increased injectivity and CO2 breakthrough time (Fig. 3.10).

The results in Table 3.6 confirm that the model chosen for the optimization task in this study is the most accurate. This is consistent with Min et al. (2011) because they also showed that incorporating PP in the input data instead of petrophysical properties increases the model accuracy. These results show that Models B and C can replicate the full physics model and be used in the well placement optimization problem. On the other hand, building a model with less informative features can result in sub-optimal solutions.

4. Discussion

This study developed an ANN-based proxy model to replace the numerical simulator in the well placement optimization workflow in the GCS. We have demonstrated that the developed proxy model can be deployed reliably by optimizing the CO_2 injection and brine production wells in the saline aquifer in the Pohang basin.

The Base Model exhibited a strong predictive performance among the developed models, with an R^2 of 0.92. This can be attributed to the effective use of the available input data despite lacking informative

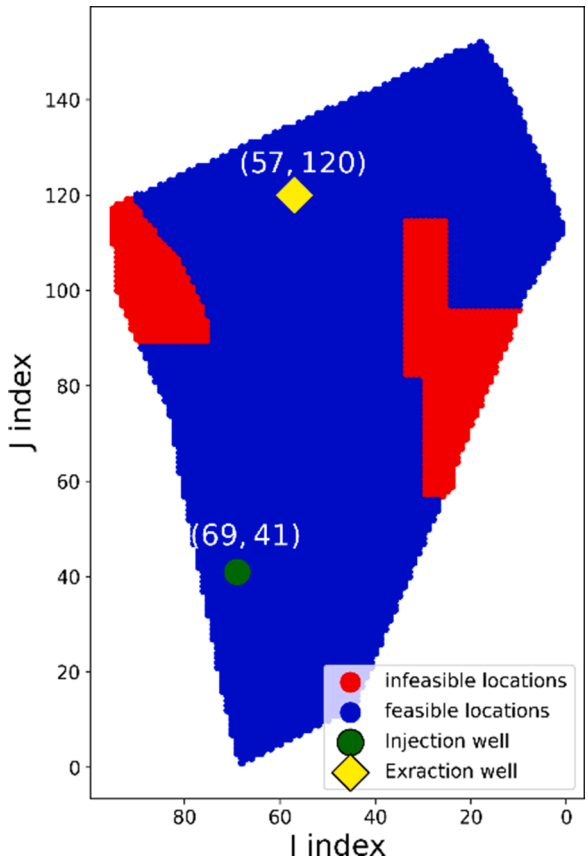


Fig. 3.8. Optimal well locations obtained by Case 3.

Table 3.5
Summary of optimization results for each scheme.

Scheme	Injection (grid cells)	Extraction (grid cells)	Cum. CO_2 (1E + 8 m^3)	Number of simulation run
Case 1	70, 42	56, 120	1.9013	622
Case 2	70, 42	54, 121	1.9002	120
Case 3	69, 41	57, 120	1.9021	141

Table 3.6
Optimal cumulative CO_2 obtained using the Base, Model A, and Model B.

Model	Optimal cum. CO_2 (m^3)
Base Model	1.8324E + 08
Model B	1.9002E + 08
Model C	1.8989E + 08

features such as permeability and porosity. On the other hand, Model A showed poor predictive performance with an R^2 of 0.48, which can be attributed to the increased dimensionality of the input features. This is because after adding the well block permeability and porosity values (corresponding to 24 variables for each well), the number of input features increased to 55 in Model A. The performances of these models may have been improved by increasing the training data, which was undesirable from the perspective of computational efficiency.

By contrast, Models B and C had superior performances on both testing data and optimization tasks. The optimal well locations obtained from Models B and C were in excellent agreement with the simulator-based optimization outcomes. This can be because the well productivity potential and the arithmetic average of the well block permeability and porosity values were used as input data to represent the well block permeability and porosity values in Models B and C, respectively. Therefore, 24 variables for each well were represented, with only one variable in Model B and two in Model C. This representation improved the performance of both models and addressed the issues of the Base Model (lack of informative feature) and Model A (high dimensionality).

Based on the different performances of the developed models, this study indicates that when developing a prediction model for a heterogeneous formation, it is essential to include the most informative features (e.g., permeability and porosity) and maintain the dimensionality of the problem tractable. This is consistent with Nwachukwu et al. (2018) and Min et al. (2011).

Updating the ANN model during the optimization task with data points close to the optimal solutions improved the accuracy of the ANN prediction in the newly discovered area of the search space. The reason for increased accuracy is the increased representation of near-optimal data points in the training data. This accords with Golzari et al. (2015) and Salehian et al. (2022), who observed similar benefits in their respective studies.

A single realization of the geological model was considered for testing the proposed optimization workflow. However, the subsurface will be uncertain in practice requiring the consideration of multiple realizations. The methodology presented in this paper can be applied easily to optimize the expected objective function over multiple realizations. This will require training data set to be prepared for multiple realizations.

Several authors have considered uncertainty in proxy-based optimization in the hydrocarbon industry (e.g., Kim et al., 2020; Nwachukwu et al., 2018). Nwachukwu et al. (2018) conducted a joint optimization of well placement and controls in an oil field considering uncertainty. According to their results, the number of required simulation runs to find the optimal solution with proxy-based optimization was approximately 880. In contrast, the optimal solution obtained from the simulator-based optimization was 9300. The uncertainty can be incorporated easily in well placement optimization in the context of GCS. Consistent with previous studies, the reduction in the required simulation runs will be more pronounced when accounting for uncertainty than the single geological realization. In this sense, the results obtained by considering a single geological realization may be conservative.

Although the developed model can evaluate the well placement scenarios with reasonable accuracy, there are some limitations to this approach. Data-driven models depend strongly on the quality of input data because they do not honor the underlying physics (Shokouhi et al., 2021). The accuracy of the optimal solutions will degrade if the input data is not representative of the whole search space. Thus, the training data is paramount for the model to predict accurately (Chollet, 2017).

Another limitation of the developed proxy model is that it can only be applied to the reservoir used for training data preparation. In addition, it can only predict the behavior of a dynamic system within the bounds of the training data, making it case-specific. For example, two wells were employed in this study, considering constant operating conditions. If one more well was added or the operating conditions were changed, it would require preparing a new training dataset to train the model. Thus, the training data set should be prepared based on the objective.

In this study, the objective function was the cumulative CO_2 storage at the end of the injection period. It may be of interest to consider other objective functions. For example, brine production will be expensive to apply in field operations. Thus, exploring cost-saving options will be helpful when dealing with brine production. This could be achieved by treating the bi-objective optimization, which aims to minimize the extracted brine amount and maximize the CO_2 stored. A proxy model that can predict the cumulative CO_2 injection and brine extraction can be developed for bi-objective optimization.

Future work may also account for more general well scenarios. In this study, only vertical wells were considered limiting the number of wells to two. Optimization can be generalized by including the type (i.e.,

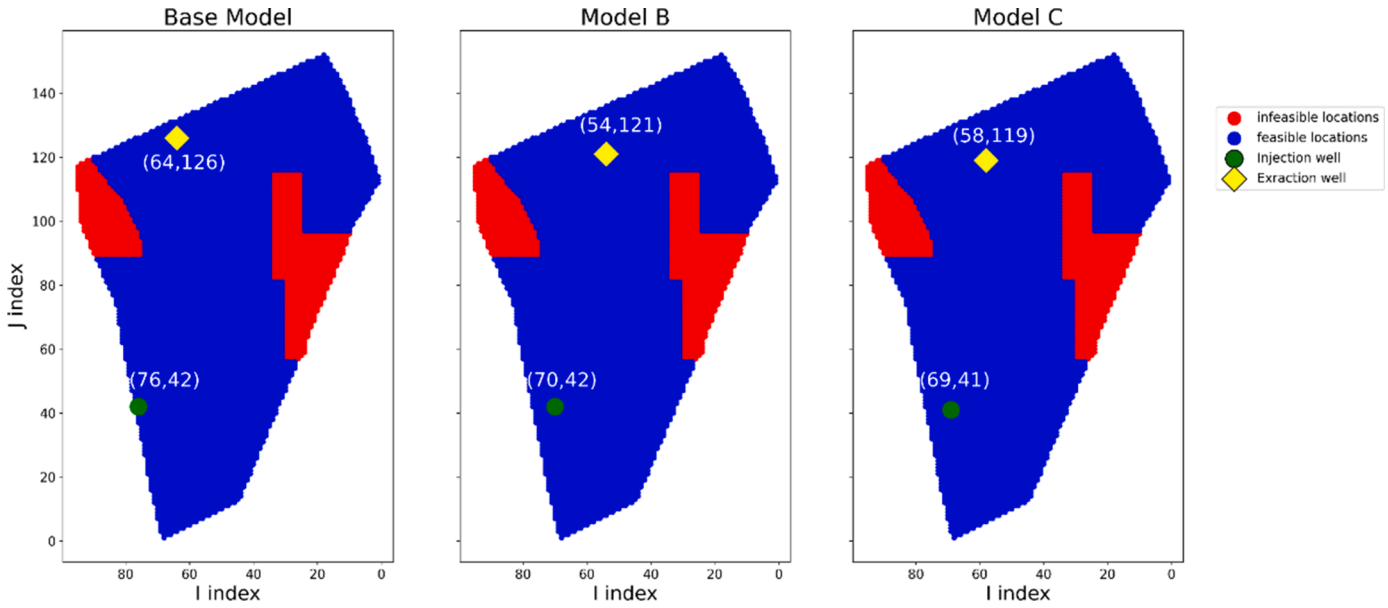


Fig. 3.9. Optimal well locations obtained using Base Model, Model B and C.

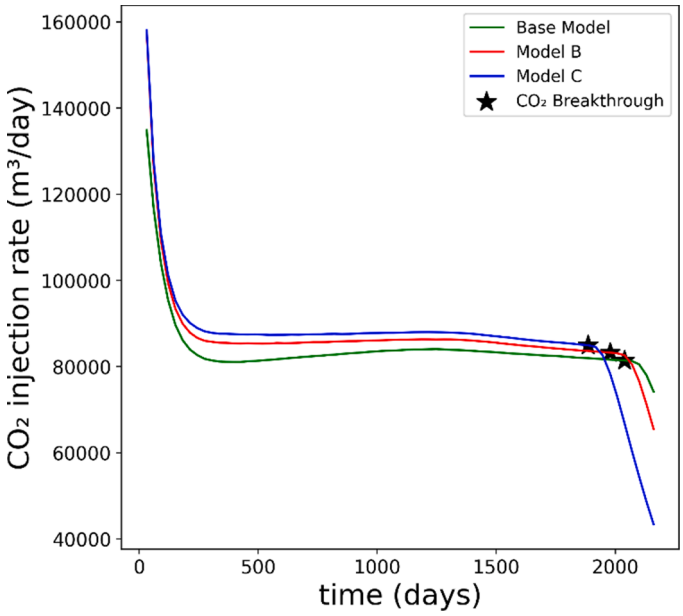


Fig. 3.10. CO_2 injection rate for the optimal well placements obtained by Base, Model B and Model C.

horizontal or multilateral) and the total number of wells. In addition, this study can be expanded to a joint optimization study of well placement and operating conditions. Finally, the proxy model developed in this study is based on an ANN, which is fully data-driven. Investigating the physics-informed deep learning models would be useful for the optimization problem.

5. Conclusions

This work presented the ANN-based optimization for CO_2 injection and brine production well placement problem in the GCS. The proposed optimization framework was evaluated in the upper aquifer of the Pohang basin. Four different ANN models with different input features were constructed. The model that yielded the best performance on the test data was selected for deployment. The GA was employed for both tuning the hyperparameters of the proxy model and for the ANN-based optimization.

The results were validated by conducting the reservoir simulator-based optimization using the DECE optimization algorithm. The simulator-based optimization required 622 simulation runs to converge to the optimum. On the other hand, ANN-based optimization achieved a comparable solution with only 120 simulation runs, reducing the number of simulation runs by 80.7%. The following are the conclusions drawn from this study:

1) ANN-based proxy is well suited for the optimization problem because of its accurate and fast predictions. Thousands of objective function evaluations can be achieved in seconds. ANN offers an alternative for tasks requiring many simulation runs.
2) Selection of the most informative input feature is paramount for optimization study as irrelevant or non-informative features can result in sub-optimal solutions.
3) Instead of using the permeability and porosity of well blocks individually as input to the model to capture the effects of heterogeneity, the inclusion of PP or average permeability and porosity along the well as input increased the accuracy of the ANN model. PP and the average permeability and porosity carry important information on the flow and storage capacity while reducing the dimensionality of the input features.
4) Optimization based on the proxy model reduced the actual numerical simulation runs significantly, reaching comparable optimal solutions. Updating the proxy model with near-optimal solutions increased the accuracy of the predictions during the optimization.

CRediT authorship contribution statement

Kudrat Musayev: Investigation, Methodology, Software, Visualization, Writing – original draft, Formal analysis. **Hyundon Shin:** Funding acquisition, Resources, Validation, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Viet Nguyen-Le:** Formal analysis, Investigation, Validation, Visualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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